Laboratory Guide 1: Introduction to Qiskit

From Theory to Practice: First Steps with Quantum Circuits

Course: Quantum Computing Foundation

1 Introduction and Objectives

This laboratory serves as the bridge between the theoretical concepts of single-qubit states and their practical implementation using the IBM Quantum Platform.

Prerequisites:

- Lesson QC03: *The Qubit and the Bloch Sphere*
- Lesson QC04: *Single-qubit Gates*
- Lab 0: *Environment Setup*

Learning Objectives:

1. Visualize quantum states on the Bloch Sphere and Q-Sphere.
2. Implement basic gates (X, H, Z) and observe their effect on probability amplitudes.
3. Run a quantum circuit using Qiskit and interpret "Shot Noise" in measurement results.
4. Demonstrate quantum interference experimentally.

Important: Visualization Note

Bloch Sphere vs. Q-Sphere
The lectures use the **Bloch Sphere** (a single vector arrow) to represent qubits. However, the IBM Composer uses the **Q-Sphere**, which represents states as **dots** on the surface.

- **Dot Size:** Represents probability magnitude ($|\alpha|^2$).
- **Dot Color:** Represents phase (Blue $= +$, Red $= -$).

Keep this difference in mind as you proceed.

2 Part A: Visual Intuition (IBM Quantum Composer)

Platform: IBM Quantum Composer (Drag-and-Drop Interface)

2.1 Exercise 1: The Bit Flip (Pauli-X Gate)

Context: Imagine a light switch. To turn it from OFF (0) to ON (1), you apply a force. The X Gate is that force. In **QC04**, we defined: $X|0\rangle = |1\rangle$.

Student Activity

Action: Drag the `X` gate onto the qubit line (q_0) initialized at $|0\rangle$.
Expected Effect:

- **Theory (Bloch):** The vector rotates 180 degrees to the South Pole ($|1\rangle$).
- **Probabilities:** The measurement probability for outcome "1" becomes 100%.

IBM Composer View (Q-Sphere)

What you see on screen: A single dot at the **Bottom** (South Pole).
Color: **Blue** (Phase 0).

2.2 Exercise 2: Creating Superposition (Hadamard Gate)

Context: Imagine spinning a coin. While spinning, it is a mix of Heads and Tails. The H Gate "spins" the qubit into superposition: $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$.

Student Activity

Action: Replace the X gate with an `H` gate.
Expected Effect:

- **Theory (Bloch):** The vector points to the Equator (X-axis).
- **Probabilities:** 50% for $|0\rangle$ and 50% for $|1\rangle$.

IBM Composer View (Q-Sphere)

What you see on screen: Two Dots (one at Top, one at Bottom).
Color: Both represent positive coefficients, so both are **Blue**.

2.3 Exercise 3: The "Invisible" Phase (Pauli-Z Gate)

Context: The Z gate creates the state $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$. The minus sign is a phase change, invisible to standard probability measurement.

Student Activity

Action: Place a `Z` gate immediately after the `H` gate.
Expected Effect:

- **Theory (Bloch):** The vector rotates along the equator to the negative X-axis.
- **Probabilities: Unchanged** (50/50).

IBM Composer View (Q-Sphere)

What you see on screen: Still two dots, but look at the colors!
Top Dot ($|0\rangle$): **Blue**.
Bottom Dot ($|1\rangle$): **Red**.
The Red color visualizes the negative sign (phase π) introduced by the Z gate.

3 Part B: Coding & Simulation (Qiskit)

3.1 Exercise 4: Measurement and Shot Noise

Coding Task

```python
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

qc = QuantumCircuit(1, 1)
qc.h(0)             # Create superposition
qc.measure(0, 0)    # Collapse and measure

# Execute with limited shots
backend = AerSimulator()
job = backend.run(qc, shots=50)
counts = job.result().get_counts()
print(f"Results for 50 shots: {counts}")
```

Analysis: Notice that for 50 shots, you rarely get exactly 25/25. This variance is the "shot noise". Try increasing `shots` to 10000 in the notebook to see the distribution stabilize.

3.2 Exercise 5: Quantum Interference (The H-Z-H Sequence)

Coding Task

```python
qc = QuantumCircuit(1, 1)
qc.h(0)    # Step 1: Superposition (|+>)
qc.z(0)    # Step 2: Phase Flip (|->)
qc.h(0)    # Step 3: Interference
qc.measure(0, 0)

job = AerSimulator().run(qc, shots=1000)
print(job.result().get_counts())
```

Expected Result: 100% state $|1\rangle$.
Why? Even though Z was invisible in the probability chart of Exercise 3, the interference in Step 3 converts that phase difference into a measurable bit flip.

4 Part C: Assessment (Quiz)

Challenge yourself! Connect the visual results with the linear algebra from Lesson QC04.

Q1. Math Challenge: The Action of H on $|1\rangle$
In Exercise 2, we applied H to $|0\rangle$ and got $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$. If we initialize the qubit to $|1\rangle$ (by adding an X gate first) and *then* apply H, what is the resulting state vector?

A) $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$ (Two Blue dots)

B) $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$ (One Blue dot, One Red dot)

C) $\begin{pmatrix}0\\1\end{pmatrix}$ (One Blue dot at South)

Q2. Physics Challenge: Explaining H-Z-H

In Exercise 5, the circuit $H \to Z \to H$ acting on $|0\rangle$ produced the deterministic output $|1\rangle$. Which physical principle explains why the output is not random (50/50)?

A) **Decoherence:** The Z gate destroyed the superposition.

B) **Constructive/Destructive Interference:** The Z gate introduced a phase shift that caused the $|0\rangle$ terms to cancel out and the $|1\rangle$ terms to sum up during the second H gate.

C) **Measurement Collapse:** The Z gate performed a hidden measurement.

Q3. Visualization Challenge

You are given a qubit in the state $|\Psi\rangle = \sqrt{0.3}|0\rangle - \sqrt{0.7}|1\rangle$. If you visualize this on the IBM Composer Q-Sphere, what will you see?

A) A small Blue dot at North and a larger Red dot at South.

B) A large Blue dot at North and a small Blue dot at South.

C) Two equal-sized Red dots.

Solutions: Q1: B (Recall $H|1\rangle = |-\rangle$); Q2: B (Interference is key); Q3: A (Square roots determine size, minus sign determines Red color).